

1. Recurrence

$$T(n) = 5T\left(\frac{n}{7}\right) + 3n^2$$

$$a = 5, b = 7, f(n) = 3n^2 \rightarrow \Theta(n^2)$$

$$n^{\log_7 5} = n^{0.83} \rightarrow n^2 > n^{0.83} \text{ (case 3)}$$

$$T(n) = \Theta(n^2)$$

2. Master Theorem

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$$\text{Case 1: } f(n) < n^{\log_b a} \rightarrow T(n) = \Theta(n^{\log_b a})$$

$$\text{Case 2: } f(n) \approx n^{\log_b a} \rightarrow T(n) = \Theta(n^{\log_b a} \log n)$$

$$\text{Case 3: } f(n) > n^{\log_b a} \rightarrow T(n) = \Theta(f(n))$$

a. $T(n) = 5T\left(\frac{n}{9}\right) + n^7$

$$a = 5, b = 9, f(n) = n^7$$

$$n^{\log_9 5} = n^{0.73} \rightarrow n^7 > n^{0.73} \text{ (case 3)}$$

$$T(n) = \Theta(n^7)$$

b. $T(n) = 3T\left(\frac{n}{3}\right) + 7n$

$$a = 3, b = 3, f(n) = 7n \rightarrow \Theta(n)$$

$$n^{\log_3 3} = n^1 \rightarrow n = n \text{ (case 2)}$$

$$T(n) = \Theta(n \log n)$$

c. $T(n) = 15T\left(\frac{n}{4}\right) + 100$

$$a = 15, b = 4, f(n) = 100 \rightarrow \Theta(1)$$

$$n^{\log_4 15} = n^{1.95} \rightarrow n^{1.95} > 1 \text{ (case 1)}$$

$$T(n) = \Theta(n^{\log_4 15})$$

d. $T(n) = 9T\left(\frac{n}{3}\right) + n^2$

$$a = 9, b = 3, f(n) = n^2$$

$$n^{\log_3 9} = n^2 \rightarrow n^2 = n^2 \text{ (case 2)}$$

$$T(n) = \Theta(n^2 \log n)$$

e. $T(n) = 6T\left(\frac{n}{5}\right) + n^3$

$$a = 6, b = 5, f(n) = n^3$$

$$n^{\log_5 6} = n^{1.11} \rightarrow n^3 > n^{1.11} \text{ (case 3)}$$

$$T(n) = \Theta(n^3)$$

3. Radix Sort

input = [BOY,RIS,GUA,ATL,SUC,CES,QUI,PUT,ANT,VAU,HUI,VAL,SAN,SAN,CAQ,VIC,CAL,LA_GUA,
TOL,META,CAU,BOL,AMA,MAG,CHO,NSA,GUA,CAS,COR]

1ST PASS:

CHAR	LAST LETTER
A	GUA, LA_GUA, META, AMA, NSA, GUA
B	
C	SUC, VIC
G	MAG
H	
I	QUI, HUI
L	ATL, VAL, CAL, TOL, BOL
M	
N	SAN, SAN
O	CHO
P	
Q	CAQ
R	COR
S	RIS, CES, CAS
T	PUT, ANT
U	VAU, CAU
V	
Y	BOY

Order: [GUA, LA_GUA, META, AMA, NSA, GUA, SUC, VIC, MAG, QUI, HUI, ATL, VAL, CAL, TOL,
BOL, SAN, SAN, CHO, CAQ, COR, RIS, CES, CAS, PUT, ANT, VAU, CAU, BOY]

2ND PASS:

CHAR	2 ND LETTER
A	LA_GUA, MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU
E	META, CES
H	CHO
I	VIC, RIS
M	AMA
N	ANT
O	TOL, BOL, COR, BOY
S	NSA
T	ATL
U	GUA, GUA, SUC, QUI, HUI, PUT

ORDER: [LA_GUA, MAG, VAL, CAL, SAN, SAN, CAQ, CAS, VAU, CAU, META, CES, CHO, VIC, RIS,
AMA, ANT, TOL, BOL, COR, BOY, NSA, ATL, GUA, GUA, SUC, QUI, HUI, PUT]

LAST PASS:

CHAR	1 ST LETTER
A	AMA, ANT, ATL
B	BOL, BOY
C	CAL, CAQ, CAS, CAU, CES, CHO, COR
G	GUA, GUA
H	HUI

L	LA_GUA
M	MAG, META
N	NSA
P	PUT
Q	QUI
R	RIS
S	SAN, SAN, SUC
T	TOL
V	VAL, VAU, VIC

**ORDER: [AMA, ANT, ATL, BOL, BOY, CAL, CAQ, CAS, CAU, CES, CHO, COR, GUA, GUA, HUI,
LA_GUA, MAG, META, NSA, PUT, QUI, RIS, SAN, SAN, SUC, TOL, VAL, VAU, VIC]**

7. COMPLEXITY ANALYSIS

For the Lexicographic Radix sort, The time complexity is $O(n * m)$ where n is the number of strings, and m is the maximum string string length, as each string is processed once and placed into a bucket before they're copied back out from the first bucket to the last. The space complexity is $O(n)$, as the buckets together store the same n strings from the array.

For the palindrome triplets algorithm, the Time complexity is $O(n^4)$, where n is the number of strings in the input array, as each for loop iterates n times and if the string concatenates all the strings from the input, the while loop will iterate $n/2$ times, which rounds to $O(n)$. The space complexity is $O(n^3)$, as in the worst case, all triplet combinations of the input strings produces a palindrome and will be stored in the hashMap.